Same Output, Different Process: Ornstein's $\bar{d}$-Distance as a Computational-Equivalence Metric for Model Diffing

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Abstract

When are two neural networks *the same computation*? Existing model-comparison tools answer at the level of static representation geometry (CKA, SVCCA, crosscoders, the Platonic hypothesis) or of the conjugacy of a *deterministic linearized* system (Dynamical Similarity Analysis, DSA). Both levels can be blind to the distinction between processes: two models can emit statistically identical symbols, occupy indistinguishable representation geometries, and share a linearized flow, while implementing *different stochastic processes*. We introduce the first application of Ornstein's $\bar{d}$-distance — a nonparametric metric on stationary processes from ergodic theory, estimated by optimal transport between empirical n-block distributions under Hamming cost — to model comparison, on deliberately low-entropy symbolic readouts where the estimator is consistent. In a pre-registered study on GRUs trained to emit hidden-Markov processes, we constructed twins of each base model (noise-injected, distilled, pruned, recoded) that preserve its output by every static measure (unigram total variation ≤ 0.0076, next-symbol cross-entropy within 1.6%). A 50%-pruned, fine-tuned twin with indistinguishable outputs (CKA $= 1.000 \pm 0.000$; DSA inside its own null range) is separated from its base by $\bar{d}$ at $18.0 \pm 12.1\times$ the same-process floor (5/5 seeds, $\geq 2.6\times$). Independent training seeds — called identical by CKA (1.000 ± 0.000) and DSA (0.080 ± 0.014) — differ by $16.4 \pm 6.1\times$ floor; sweeping the prune fraction from 0.1 to 0.9 keeps DSA at its null throughout while $\bar{d}$ separates at every fraction. Conversely $\bar{d}$ correctly certifies GRU distillation as process-preserving at the output level (transformer distillation is not: students separate on every seed of two grammars, while DSA rates them as *more* different than models of entirely different tasks), and a different-computation control with *identical* entropy rate and unigram marginal is separated at $194.9 \pm 21.3\times$ floor while DSA cannot distinguish it from its own null. We report the pre-registered gate verdict (which failed as written), the two disclosed analysis amendments that repair it, every same-process floor, and the block-length consistency wall $n \lesssim \log_2 N/\hat{h}$ that bounds all claims. Per Ornstein–Weiss, $\bar{d}$ is not a general isomorphism detector; the claim is a process-level distance within a fixed readout — and at that level it sees changes every static and deterministic-conjugacy metric we tested misses.

1 Introduction

Model comparison is becoming a load-bearing operation in deep learning. We compare a fine-tune against its base to ask what changed [8, 13], a distilled or pruned model against its teacher to ask what was lost, and independently trained networks against each other to ask whether they converge on shared solutions [7]. The tools used for these questions are, almost without exception, *static*: CKA [10], SVCCA [20], crosscoder feature diffing [8], and their relatives compare the geometry of representations at a single time slice. The survey of Klabunde et al. [9] taxonomizes dozens of similarity measures; none is a metric on the *stochastic process* a model's computation unfolds in time.

The strongest existing move beyond static geometry is Dynamical Similarity Analysis [DSA; 19], which delay-embeds trajectories, fits a reduced-rank linear propagator (DMD), and measures conjugacy of the fitted operators over orthogonal changes of basis. DSA asks: *do two systems have the same deterministic linearized flow?* That is a dynamical question, but not a process-level one: it is by construction insensitive to how a computation is *randomized* — and, as we show, its estimation pipeline can be both blind to genuinely different symbolic processes and loudly triggered by process noise that leaves the deterministic flow untouched.

Ergodic theory owns the missing notion. Ornstein's $\bar{d}$-distance [5, 16, 17] is a metric on stationary processes over a common alphabet: $\bar{d}(\mu, \nu)$ is the minimal long-run fraction of symbol mismatches over all stationary couplings of the two processes. It metrizes *process* equivalence — $\bar{d} = 0$ iff the two process laws coincide — and it is exactly the tool for "same one-time statistics, different dynamics": two processes can share every marginal, and even every spectral and entropy invariant, while sitting at large $\bar{d}$.

The ceiling, stated first. Before claiming anything, we state what cannot be claimed. Ornstein and Weiss [18] proved that entropy is essentially the *only* finitely observable isomorphism invariant of stationary processes: no finite-data procedure that is constant on isomorphism classes can compute anything beyond a function of entropy. $\bar{d}$ escapes this theorem only because it is *not* an isomorphism invariant — it is a distance on processes *as coded in a fixed alphabet*. Consequently, nothing in this paper claims to detect "the same computation up to recoding" in general. Our claim is strictly: *within a fixed, deliberately low-entropy symbolic readout, $\bar{d}$ estimates a process-level distance that separates same-output/different-process model pairs which static and deterministic-conjugacy metrics do not separate.* Relatedly, consistent estimation of $\bar{d}$ itself is delicate [21]: the empirical n-block estimator is consistent only for block lengths $n \lesssim \log N/h$ [11], and it degrades *silently* beyond that wall. Both constraints are built into our protocol as hard guards (§4), and one of our headline plots (Fig. 3) is precisely the demonstration that our separations live below the wall and our nulls stay at the floor.

Contributions.

1. **A new class of model-comparison metric.** The first application (to our knowledge, and after a deliberate scoop-check) of Ornstein's $\bar{d}$ — or any ergodic-theoretic process metric — to comparing neural networks. The estimator is exact optimal transport between empirical n-block distributions under normalized Hamming cost, with same-process floors and a consistency-wall guard inherited from its use on chaotic systems [6].

2. **An existence proof.** On GRUs emitting hidden-Markov processes, a 50%-pruned and fine-tuned twin whose outputs are indistinguishable from its base (unigram TV ≤ 0.008; cross-entropy recovered to the base level; CKA $= 1.000 \pm 0.000$; DSA $= 0.074 \pm 0.029$, inside DSA's null spread $[0.028, 0.118]$) is separated by $\bar{d}$ at $18.0 \pm 12.1\times$ its same-process floor, on every seed. Independent seeds — "the same model" to every baseline — differ at $16.4 \pm 6.1\times$ floor.

3. **Failure modes of the incumbent, in both directions.** On a control pair built to have *identical* entropy rate and unigram marginal but provably different processes (golden-mean

vs. flipped-even), $\bar{d}$ reports $194.9 \pm 21.3\times$ floor while DSA's score (0.117) is statistically inside its own null distribution. Conversely, hidden-state noise that leaves the deterministic dynamics and the emitted process nearly unchanged saturates DSA (0.963 ± 0.020) — an errors-in-variables artifact of fitting a linear propagator through process noise — while $\bar{d}$ reports a calibrated, modest process change.

4. **A certification use-case.** For distilled twins, $\bar{d}$ at the emitted readout sits at the same-process floor: distillation here *did* preserve the output process, and $\bar{d}$ certifies that — while the belief-state readout reveals the students' internal predictive processes are *not* identical to the teacher's.

5. **Registered-report honesty.** The full pre-registered design (gate, kill-conditions, thresholds) was committed before any results; the gate *failed as written* due to two identifiable artifacts of our own gate arithmetic, and we report both the pre-registered verdict and the disclosed amendments side by side, with all raw curves.

2 Background: $\bar{d}$, its estimator, and its limits

2.1 The distance

Let μ, ν be stationary ergodic processes on a finite alphabet $\mathcal{A}$, $|\mathcal{A}| = m$. Ornstein's $\bar{d}$-distance is

$$\bar{d}(\mu, \nu) \;=\; \inf_{\lambda \in J(\mu,\nu)} \; \mathbb{E}_\lambda[\mathbf{1}\{X_0 \neq Y_0\}]\,, \tag{1}$$

the infimum over stationary couplings $J(\mu, \nu)$ of the per-symbol disagreement rate. Equivalently [5], $\bar{d}(\mu, \nu) = \lim_n \bar{d}_n(\mu, \nu)$ where $\bar{d}_n$ is the optimal-transport distance between the n-block marginals μ_n, ν_n on $\mathcal{A}^n$ under normalized Hamming cost $c(u, v) = \frac{1}{n}\#\{i : u_i \neq v_i\}$. The sequence $\bar{d}_n$ is non-decreasing, so every $\bar{d}_n$ is a *lower bound* on $\bar{d}$. Two properties make $\bar{d}$ the right notion for "same output, different process": (i) $\bar{d}_1$ is exactly the total variation between the one-symbol marginals — the "invariant measure" of the readout — so the split between $\bar{d}_1$ and $\bar{d}_{n\geq 2}$ is precisely the split between static and dynamical difference; (ii) $\bar{d}(\mu, \nu) = 0$ iff $\mu = \nu$ as process laws, i.e. iff *all* temporal statistics agree.

2.2 The estimator

We reuse the estimator of Howe [6], built for evaluating surrogates of chaotic systems: form empirical n-block distributions from N observed symbols (pooled over independent chains, blocks never crossing chain boundaries) and solve the discrete OT problem exactly with a network simplex [4]. When the observed block support exceeds 4096 types, we switch to a sampled regime (2000 blocks per side, 4 repeats). Two honesty devices are inherited and used throughout:

Same-process floors. Empirical OT between two independent samples of the *same* process is strictly positive at finite N. Every pair estimate $\bar{d}_n(A, B)$ is reported against $\max\{\bar{d}_n(A, A'), \bar{d}_n(B, B')\}$, where A' and B' are independent generation runs of the same models, computed with the identical regime and budget. Only the excess over the floor is treated as signal.

The consistency wall. Marton and Shields [11] showed the empirical n-block distribution is a consistent estimator (in $\bar{d}$) only for $n \lesssim \log_2 N/h$, with h the process entropy rate in bits; beyond that the estimate degrades without warning. We estimate $\hat{h}$ by conditional block entropies under an undersampling guard (LZ78 as cross-check), compute the wall $k_{\text{wall}} = \log_2 N/\hat{h}$ per stream, and *restrict every claim to block lengths below it.* This is the constraint that kills the naive idea of running $\bar{d}$ on raw activation streams: at $h \approx 2$–4 bits/step, affordable budgets buy only $n \approx 4$–10 — hence the low-entropy readouts below.

As an independent certificate, a Fano-type inequality [cf. 23] gives $|h(\mu) - h(\nu)| \leq H_{\text{b}}(\bar{d}) + \bar{d}\log_2(m-1)$, so an entropy-rate gap lower-bounds $\bar{d}$ without any OT machinery; we report it where informative.

2.3 What may and may not be claimed

$\bar{d}$ is alphabet-bound: it compares processes *as symbolized.* A model and a recoded copy of itself have $\bar{d} = 0$ at any behavioral readout (our null pair tests exactly this), but $\bar{d}$ says nothing about whether two *internally different* models are isomorphic as abstract dynamical systems — by Ornstein and Weiss [18], no finite-data quantity beyond entropy could. Our claims are therefore always of the form: *at this readout, the processes differ by at least this much* (lower-bound semantics, floors subtracted), or *at this readout and budget, no difference above the floor is detectable.*

3 Related work

DSA [19] is the competitor and the explicit foil: it measures whether two networks "perform the same computation with equivalent dynamics" via delay-embedded reduced-rank DMD plus a Procrustes alignment over orthogonal transformations (we use the authors' public implementation and their angular score; accelerated variants exist [2]). DSA tests conjugacy of a *deterministic linearized* system; $\bar{d}$ tests equality of the *full nonparametric stochastic* process at a readout. The two disagree exactly where those notions come apart, and §5 demonstrates both directions of disagreement empirically.

OT between noisy neural dynamics. Nejatbakhsh et al. [14] is the closest work mechanically — optimal transport between stochastic neural processes, with causal couplings — but its distances are computed on Gaussian/second-order statistics of trajectories in $\mathbb{R}^d$. $\bar{d}$ is nonparametric over symbol sequences: it sees all orders of temporal dependence, at the price of requiring a symbolic readout.

Belief-state geometry. Shai et al. [22] show transformers trained on hidden-Markov emissions linearly represent the belief simplex of the generator — characterizing the process *one* model computes against ground truth. We use their generator style (and their framing motivates our quantized-belief readout), but our object is a *metric between two models.*

Crosscoders and model diffing. Crosscoder-based diffing [3, 8] and activation-difference readouts [13] ask "what changed between base and fine-tune" at the level of static features. Our proposal is the dynamical analog of that program: diff the *process*, not the feature dictionary.

Static similarity. CKA [10], SVCCA [20], and the Platonic-convergence line [7] compare one-time geometry; Klabunde et al. [9] confirm process-level distances are absent from the current taxonomy.

Automaton extraction. L*-style and clustering-based extraction of automata from RNNs/transformers [1, 15, 24] decides discrete equivalence of extracted machines; it produces low-entropy symbolizations compatible with our estimator but no *metric* on the stochastic process. Our emitted-token readout is the degenerate, extraction-free case; plugging extracted-automaton alphabets into $\bar{d}$ is a natural extension.

4 Method

4.1 Tasks and models

We need tasks whose emitted processes are genuinely stochastic, low-entropy, and theoretically pinned. We train 1-layer GRUs (width 64) by next-symbol cross-entropy on the **golden-mean** process (GM: binary, no "11", $h = 2/3$ bits/symbol, $P(1) = 1/3$) and, for the control, the **flipped-even** process (FEVEN: binary, runs of 0s between 1s have even length). FEVEN is chosen because it has *identical* entropy rate (2/3) and *identical* unigram marginal ($P(1) = 1/3$) to GM while being a different sofic process — so no distance to it can be explained by first-order or entropy statistics. All base models train to within 0.5% of the entropy-rate optimum (val. CE 0.6635–0.6648 bits vs. $h = 0.6667$), i.e. they emit close-to-ideal samples of the target process. E2 (§7) repeats the design with 2-layer causal transformers and adds the 3-symbol MESS3 process [12, 22].

4.2 Readouts

Emitted-token readout (primary). A trained autoregressive model *is* a stationary stochastic process: run it free (sample from its own softmax), and record the emitted symbols. We generate 64 independent chains of 32 768 steps (2.1×10^6 symbols per run; two independent runs per model for floors). This readout is canonical (the task alphabet), internally coding-invariant (recoding a model cannot change it), and low-entropy ($\hat{h} \in [0.664, 0.673]$ bits, wall $k_{\text{wall}} \approx 31.4$).

Quantized-belief readout (secondary). During the same free runs we record the model's predictive probability $P(\text{next} = 1 \mid \text{history})$ and quantize it into 4 equal bins — a symbolization of the model's *belief process* [22] (alphabet 4, wall ≈ 15.9). It captures the conditional-probability dynamics that the emitted stream integrates out.

4.3 Constructed pairs

Per base seed (5 seeds): **recoded null** (random permutation of hidden units — provably the same function; all metrics must call "same"); **independent seed** (same task, different init); **noise twin** (i.i.d. $\mathcal{N}(0, \sigma^2)$ added to the hidden state every step, in generation and in state collection; σ^* = largest grid value keeping unigram TV ≤ 0.01 and CE within 2% — the calibration never bound, so $\sigma^* = 0.2$ with worst-case TV 0.0033 and CE +1.6%); **distilled twin** (fresh student, KL to teacher conditionals, to 0.5% of teacher CE); **pruned twin** (50% global magnitude pruning, masked fine-tuning back to base CE — achieved: pruned CE $\leq$ base CE on all seeds); **different task** (GM model vs. FEVEN model).

4.4 Metrics, side by side

For every pair we report three numbers computed on the *same* models under the *same* conditions (twin noise active everywhere): **CKA** (linear, column-centered) on hidden states over a shared task-sampled evaluation set; **DSA** (public `dsa-metric`, one pre-registered configuration: 8 delays, rank 32, angular score, 1500 iterations) on the same state trajectories; and $\bar{d}_n$ curves on the symbolic readouts with per-n same-process floors, plus $\hat{h}$, the wall, and the Fano bound. $\bar{d}_1$ doubles as the matched-marginal check.

Table 1: E0, three metrics side by side (mean ± std over 5 seeds). $\bar{d}$ columns give the amended-plateau estimate as a multiple of the same-process floor (1 = indistinguishable from the same process at this budget); "belief" is the quantized-belief readout. Every pair except the control preserves outputs: CKA ≈ 1, unigram TV = $\bar{d}_1 < 0.008$, CE within calibration bounds.

pair	CKA ↑	DSA ↓	$\bar{d}_1$ (TV)	$\bar{d}_{n^*}$/floor	belief $\bar{d}_{n^*}$/floor
recoded (null)	1.000 ± 0.000	0.055 ± 0.035	0.0002 ± 0.0002	1.0 ± 0.0	1 ± 0
independent seed	1.000 ± 0.000	0.080 ± 0.014	0.0035 ± 0.0018	16.4 ± 6.1	616 ± 384
noise twin (σ=0.2)	0.996 ± 0.001	0.963 ± 0.020	0.0009 ± 0.0002	4.3 ± 1.0	448 ± 336
distilled twin	1.000 ± 0.000	0.096 ± 0.030	0.0003 ± 0.0000	1.7 ± 0.6	16 ± 19
pruned twin (50%)	1.000 ± 0.000	0.074 ± 0.029	0.0026 ± 0.0027	18.0 ± 12.1	661 ± 635
different task (ctrl)	0.413 ± 0.029	0.117 ± 0.019	0.0053 ± 0.0022	194.9 ± 21.3	314 ± 129

4.5 Pre-registration, gate, and amendments

The full design — pairs, budgets, thresholds, and three kill conditions (K1: no separation ⇒ publish the negative; K2: separations above the wall are artifacts; K3: no isomorphism claims) — was committed to the repository (`PLAN.md`) before any experiment ran. The pre-registered E0 gate asked for a pair that is CKA-similar (≥ 0.90) *and* DSA-similar (below a null-to-control interpolation threshold) *and* $\bar{d}$-separated (plateau $\geq 2\times$ floor, 2σ across seeds, marginals matched, below the wall).

The gate failed as written, for reasons visible in the committed curves and worth reporting because they are generic pitfalls: (i) the pre-registered plateau rule $n^* = \arg\max_n[\bar{d}_n - \text{floor}_n]$ sometimes lands on sampled-regime rows ($n \geq 16$) where the finite-sample OT bias inflates *both* pair estimate and floor to $\approx 0.045 \pm 0.007$ (against full-regime floors of ≈ 0.0006 at $n = 8$) — there the criterion "$\bar{d}_{n^*} \geq 2\times$ floor" is structurally unsatisfiable even for clearly separated pairs; (ii) the DSA-similarity threshold interpolated between DSA's null mean ($\nu = 0.055$) and its different-task mean ($\kappa = 0.117$) — but these two distributions *overlap* (null spread $[0.028, 0.118]$; different-task spread $[0.092, 0.150]$), so the threshold (0.070) sits inside DSA's own null noise. Amendment 1 (committed with the results, evaluated by a separate script on the unmodified JSONs) makes the two minimal repairs: plateau candidates must individually clear $2\times$ their floor, and DSA-similar means "inside DSA's empirical null distribution" (≤ 0.125). We report pre-registered and amended verdicts side by side below; no data were re-collected.

5 E0: the existence proof

5.1 Reading the table

Validation battery. The recoded null passes everywhere: $\bar{d}$ within $2\times$ floor at every n below the wall on all seeds (emitted *and* belief readouts), CKA = 1.000, DSA small (V1). The different-task control is separated by $\bar{d}$ at $194.9 \pm 21.3\times$ floor (V2) — and note $\bar{d}_1 = 0.0053 \pm 0.0022$: by construction of FEVEN, unigram marginals and entropy rates match, so *no static or entropy statistic explains the separation*; the process structure does. CKA also sees this pair (0.413 ± 0.029); DSA does not (next paragraph).

DSA fails in both directions. Its different-task scores ($\kappa = 0.117$) lie within its own null spread ($[0.028, 0.118]$): under DSA, GM-vs-FEVEN looks like a change of basis. Its noise-twin scores (0.963 ± 0.020) are an order of magnitude above everything else — yet the noise twin's

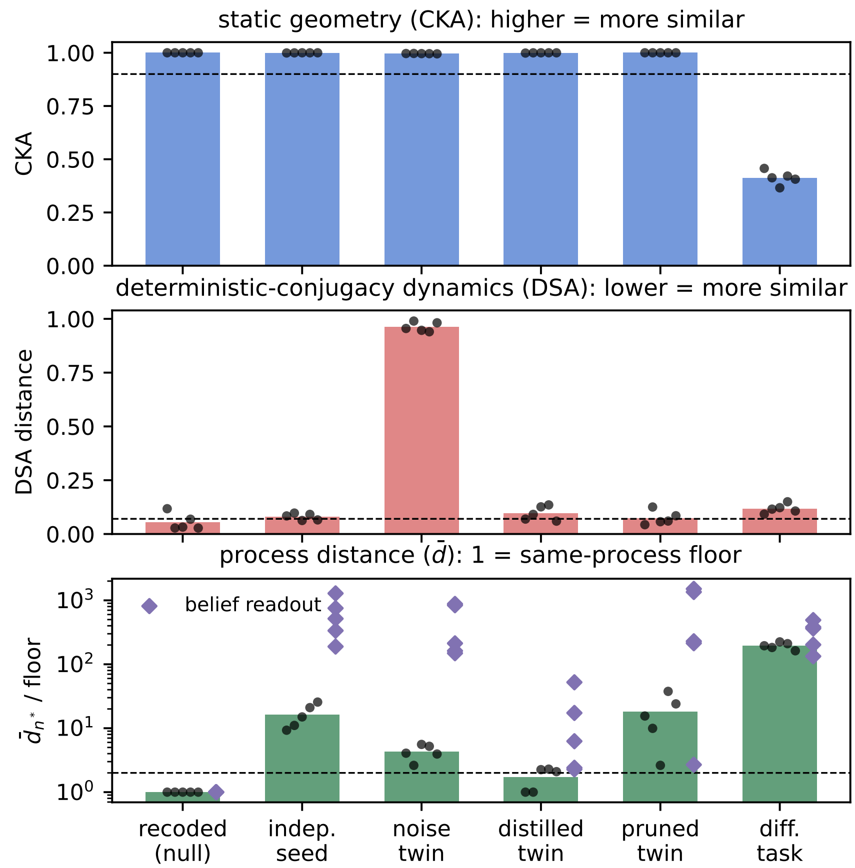

Figure 1: E0 summary. **Top:** CKA calls every same-task pair identical ($\geq 0.996 \pm 0.001$) and sees only the different-task control. **Middle:** DSA's null (recoded) spread overlaps its different-task scores — it cannot distinguish a genuinely different computation from a change of basis — while the noise twin, whose deterministic dynamics are *identical* to its base, saturates the score. **Bottom:** $\bar{d}$ as a multiple of the same-process floor (log scale; bars = emitted readout, diamonds = belief readout). The null sits at the floor; prune, independent-seed, and noise twins separate; the control separates by two orders of magnitude.

deterministic dynamics are identical to its base by construction; what DSA measures there is the errors-in-variables bias that process noise induces in a fitted linear propagator, not a dynamical difference. We stress this is not a bug in the DSA code (we use the authors' implementation and pre-registered its configuration): it is what "conjugacy of the fitted linearization" means under stochastic perturbation. DSA answers a different question, and these are the cases where the questions come apart.

The existence pair. The pruned twin has outputs indistinguishable from base (CE recovered to or below base level; TV ≤ 0.0076), CKA $= 1.000 \pm 0.000$, and DSA $= 0.074 \pm 0.029$ — inside the null spread, i.e. DSA-similar under the amended (and any reasonable) criterion. $\bar{d}$ separates it on *every* seed at 2.6×–37.7× its floor (mean 18.0 ± 12.1). The pre-registered "mean $-2\,$std > 0" inequality fails — but by effect-size heterogeneity, not by inconsistency: all five deltas are positive (sign-consistent, binomial $p = 1/32$) and the smallest is 2.6× its floor; one seed simply

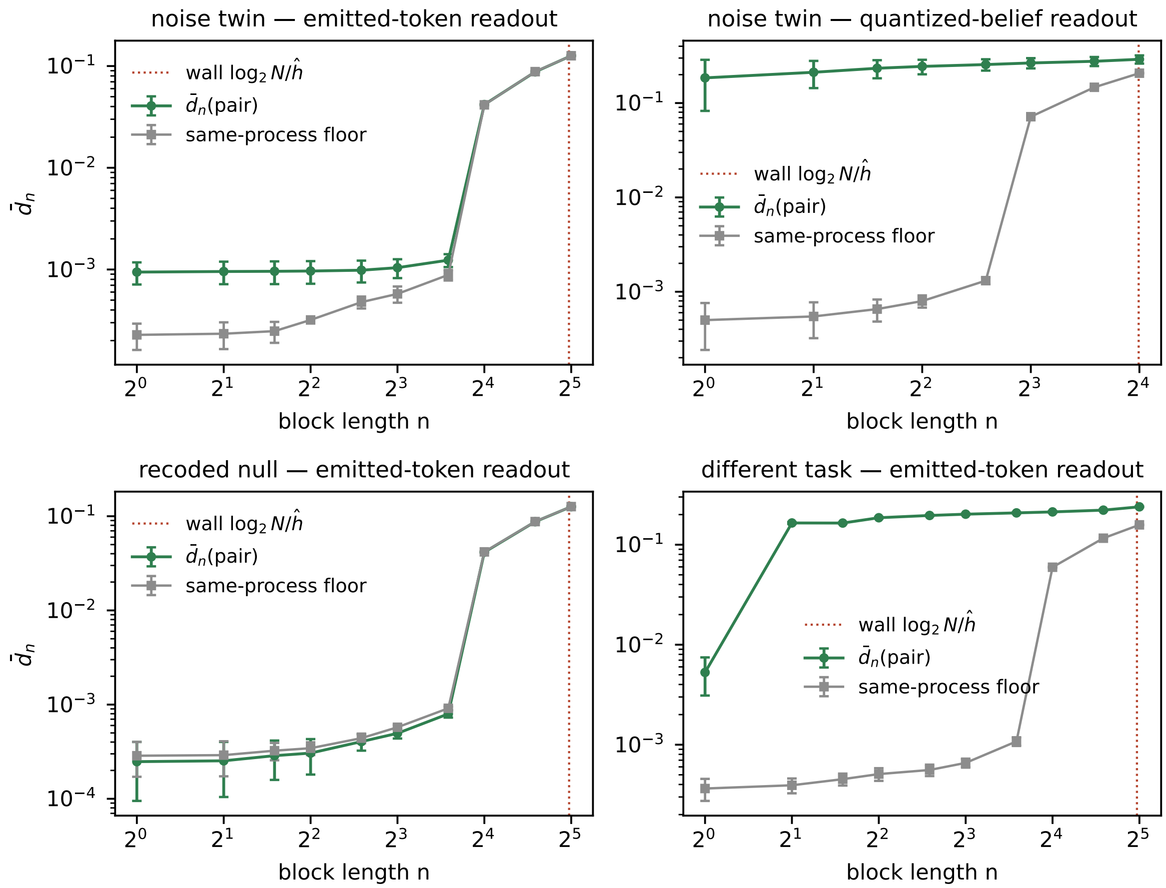

Figure 2: $\bar{d}_n$ curves (mean $\pm$ std over seeds) against same-process floors, with the consistency wall marked. **Top left:** the noise twin separates from its floor at $n \leq 12$; at $n = 16$ the estimator enters the sampled regime and pair and floor collapse onto the common OT bias — visible, not hidden. **Top right:** the same twin at the belief readout: separation from $n = 1$ on, two orders above floor. **Bottom left:** the recoded null never leaves its floor at any n (the isomorphism sanity check). **Bottom right:** the different-task control: $\bar{d}_1 \approx 0.0053 \pm 0.0022$ (unigram marginals *match* by construction), yet $\bar{d}_2 = 0.165 \pm 0.003$ — the static/dynamic split in a single curve.

changed *more*. Pruning at matched outputs is a real, $\bar{d}$-visible process change that neither CKA nor DSA registers.

Independent seeds are not the same process. The independent-seed pairs — CKA 1.000 ± 0.000, DSA 0.080 ± 0.014 (both "same") — differ at $16.4 \pm 6.1\times$ floor (emitted) and hundreds of times floor at the belief readout. Networks trained to the same near-optimal loss on the same task implement measurably different conditional processes. Any workflow that treats seed-mates as interchangeable "same computations" is, at the process level, wrong — and only the process metric among our three sees it.

The noise twin, and where the pre-registered bet failed honestly. We pre-registered noise injection as the highest-probability win on the grounds that DSA's deterministic route must be blind to it. The data falsified the second half: at the calibrated $\sigma^* = 0.2$ (chosen maximal under output preservation — the preservation constraint never bound), DSA is not blind but

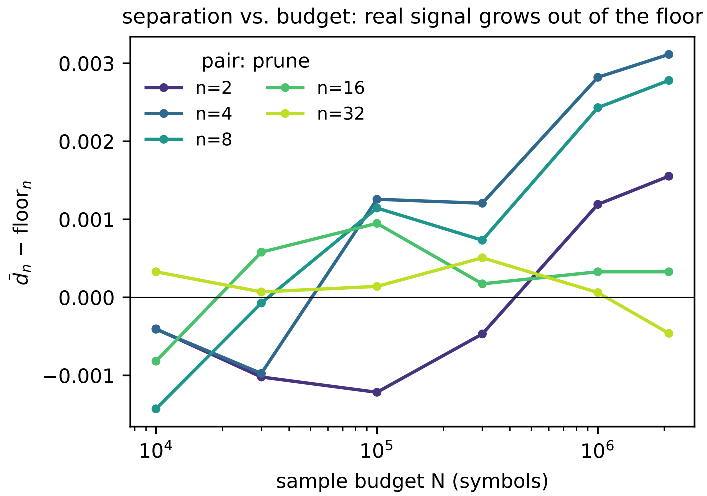

Figure 3: V3 convergence study (pruned pair, seed 0): excess $\bar{d}_n - \text{floor}_n$ vs. budget N. The separation at $n = 4$–8 grows monotonically out of the floor with budget (at full budget, $\Delta_{n=8} = 0.0028$), while at $n = 32$ — at the consistency wall — no signal accumulates ($\Delta_{n=32} = -0.0005$): the claimed separation is a stable, below-the-wall signal, not an estimation artifact (K2 respected).

saturated. $\bar{d}$ meanwhile behaves exactly as designed: marginals matched ($\bar{d}_1 = 0.0009 \pm 0.0002$), entropy rates nearly identical (Fano bound ≤ 0.00008), yet the emitted process separates at $4.3 \pm 1.0\times$ floor (all seeds, 2σ-consistent: $\Delta = 0.00072 \pm 0.00021$) and the belief process at 149–878× floor. The σ-sweep in E1 completes this picture across the grid.

Distillation: a correct null and a two-level reading. The distilled students sit at $1.7 \pm 0.6\times$ floor at the emitted readout — given the floors, that is "process preserved to measurement precision": $\bar{d}$ *certifies* the distillation, which is precisely the question one wants answered of a distillate ("did the process change or only the outputs?"). The belief readout adds nuance: students' internal predictive processes differ from the teacher's ($16 \pm 19\times$ belief floor, heterogeneous across seeds) — same emitted process, measurably different belief dynamics.

6 E1: when does $\bar{d}$ separate, and when does DSA wake up?

E1 characterizes the E0 findings along the two knobs that produced them: the noise level σ and the prune fraction, 5 seeds each, at a reduced characterization budget (Amendment 2: 8192-step runs with their own matched floors, blocks $n \leq 12$; the E0 existence claims are untouched by this).

The noise axis inverts the pre-registered bet. We pre-registered noise injection as the likeliest existence pair on the grounds that DSA's deterministic pipeline would be blind to it. The sweep shows the opposite failure: DSA leaves its null already at the smallest tested level ($\sigma = 0.01$: DSA $= 0.553 \pm 0.018$) and saturates by $\sigma = 0.05$ — a response to *noise itself*, not to the size of the process change (at $\sigma = 0$ it returns 0.000 ± 0.000 exactly, and its response to noise exceeds its response to a genuinely different computation, 0.117, several-fold). $\bar{d}$ grades instead of saturating: the belief readout separates at every $\sigma > 0$ ($\geq 74\times$ floor at all levels, already $74 \pm 39\times$ at $\sigma = 0.01$) with unigram TV ≤ 0.0015 throughout, while the emitted readout at this

Table 2: E1 sweeps (mean ± std over 5 seeds; $\bar{d}$ as multiples of same-process floors at the reduced budget). **Left:** noise twins. **Right:** pruned + fine-tuned twins; CE is the fine-tuned validation cross-entropy (base: 0.6635–0.6648; task optimum 0.6667).

σ	CKA	DSA	$\bar{d}$/fl	belief
0.0	1.000 ± 0.000	0.000 ± 0.000	1.8 ± 1.1	2 ± 1
0.01	1.000 ± 0.000	0.553 ± 0.018	1.0 ± 0.0	74 ± 39
0.02	1.000 ± 0.000	0.588 ± 0.017	1.2 ± 0.4	93 ± 71
0.05	1.000 ± 0.000	0.939 ± 0.020	1.0 ± 0.0	93 ± 67
0.1	0.999 ± 0.000	0.958 ± 0.019	2.4 ± 1.9	84 ± 47
0.2	0.996 ± 0.001	0.963 ± 0.020	2.4 ± 1.3	215 ± 220

frac	CKA	DSA	$\bar{d}$/fl	CE
0.1	1.000 ± 0.000	0.072 ± 0.012	12.4 ± 6.6	0.6637 ± 0.0005
0.3	1.000 ± 0.000	0.096 ± 0.023	9.1 ± 3.8	0.6636 ± 0.0003
0.5	1.000 ± 0.000	0.075 ± 0.012	6.3 ± 1.9	0.6635 ± 0.0001
0.7	1.000 ± 0.000	0.079 ± 0.015	4.9 ± 2.5	0.6636 ± 0.0001
0.9	1.000 ± 0.000	0.118 ± 0.025	8.2 ± 6.3	0.6646 ± 0.0004

reduced budget separates only intermittently (at most 3/5 seeds per level) — correctly reflecting that hidden noise barely perturbs the emitted process (which is precisely why these twins pass output preservation). So on the noise axis there is no window where DSA is quiet and $\bar{d}$ fires; instead, DSA fires *indiscriminately* while $\bar{d}$ resolves which level of the computation changed and by how much.

The prune axis is a disagreement window from end to end. At every fraction from 0.1 to 0.9, the fine-tuned pruned twin restores base-level outputs (CE ≤ 0.6651; TV ≤ 0.0081), CKA stays at ≥ 0.9996, DSA stays inside its null (max 0.118) — and $\bar{d}$ separates: 4.9–12.4× floor at the emitted readout, 125–397× at the belief readout. Notably even 10% pruning (plus recovery fine-tuning) yields a $\bar{d}$-visible process change.

An honest confound, bracketed. The pruned twin differs from base by pruning *and* by the recovery fine-tune; some of the separation is attributable to retraining as such. The comparison set brackets this: distilled students (fresh init, conditionals matched directly by KL) sit at $1.7 \pm 0.6\times$ floor; pruned + CE-fine-tuned twins at 4.9–12.4×; independent seeds (fresh init, CE only) at $16.4 \pm 6.1\times$. Process preservation tracks how directly the twin's training objective constrains the base model's conditionals — which is itself a $\bar{d}$-visible, and practically relevant, regularity: if you want a twin that is the *same process*, distill it; do not merely retrain it to the same loss.

7 E2: beyond toy RNNs — transformer readouts

E2 repeats the design with 2-layer causal Transformer-LMs (3 seeds; states = the final-layer residual stream; free-running generation over a sliding window, so the emitted stream is an order-255 stationary Markov process), on GM and on MESS3 [12, 22] — a 3-symbol, non-unifilar generator whose ε-machine is infinite-state, with $\hat{h} \approx 1.575$ bits, hence a much tighter wall ($k_{\text{wall}} \approx 12.1$): claims there live at $n \leq 8$. All models train to their task optimum (GM: CE ≤ 0.6656; MESS3: CE ≤ 1.5750). Twins: noise into the residual stream ($\sigma^* = 0.20$, calibration again never binding) and a distilled twin; the GM-vs-MESS3 cross pair anchors "genuinely different" for CKA/DSA.

Distillation separates, on both grammars. Transformer distillation (same architecture, fresh init, KL to teacher) is *less* process-faithful than its GRU counterpart: the students separate from their teachers on every seed — 3.9–28.1× floor on GM (CKA $= 0.975 \pm 0.003$, above the similarity line) and $9.5 \pm 2.4\times$ floor on MESS3, all within the (tight) walls. DSA is again uninformative in the direction that matters: its distill score (0.265 ± 0.006) *exceeds* its different-

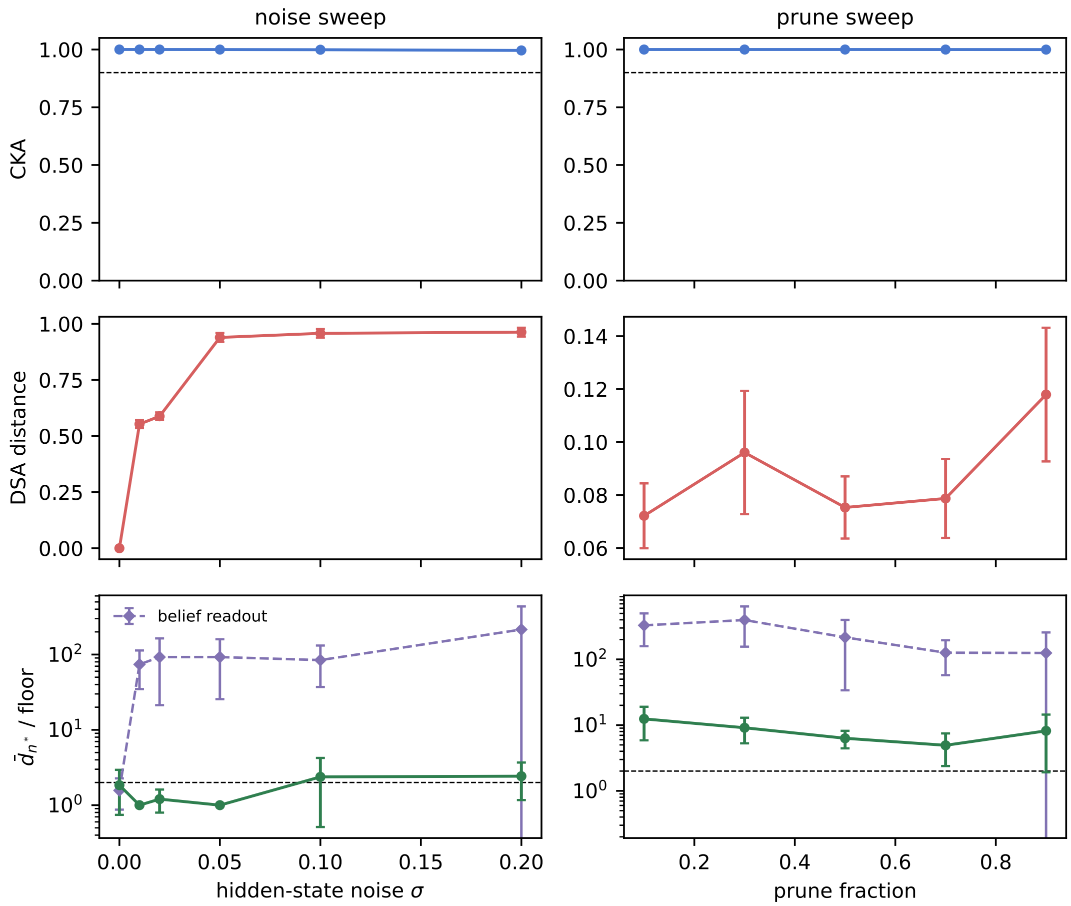

Figure 4: E1: the three metrics as functions of noise level (left column) and prune fraction (right column). Bottom row: $\bar{d}$ as a multiple of its same-process floor (log scale), emitted (circles) and belief (diamonds) readouts. DSA responds to the smallest tested noise more strongly than to a different computation, and does not respond to pruning at any fraction; $\bar{d}$'s response is graded and readout-resolved.

Table 3: E2: transformers (mean ± std over 3 seeds; $\bar{d}$ at the amended plateau, multiples of same-process floors). Cross anchor (GM vs. MESS3 models): CKA $= 0.928 \pm 0.027$, DSA $= 0.228 \pm 0.004$.

pair	CKA ↑	DSA ↓	$\bar{d}_1$ (TV)	$\bar{d}_{n^*}$/floor
GM, noise twin	0.999 ± 0.000	0.076 ± 0.007	0.0002 ± 0.0002	1.4 ± 0.6
GM, distilled twin	0.975 ± 0.003	0.265 ± 0.006	0.0041 ± 0.0019	12.1 ± 11.3
MESS3, noise twin	0.995 ± 0.001	0.121 ± 0.067	0.0014 ± 0.0004	1.0 ± 0.0
MESS3, distilled twin	0.786 ± 0.027	0.201 ± 0.005	0.0150 ± 0.0068	9.5 ± 2.4

computation anchor (0.228 ± 0.004) — under DSA, a distilled twin of the same model looks more different than a model of another task. The cross anchor also calibrates CKA here: transformers on *different* grammars score CKA 0.928 ± 0.027 — higher than some same-task distill pairs — so on this testbed neither baseline ranks "how different is the computation" correctly, while $\bar{d}$'s

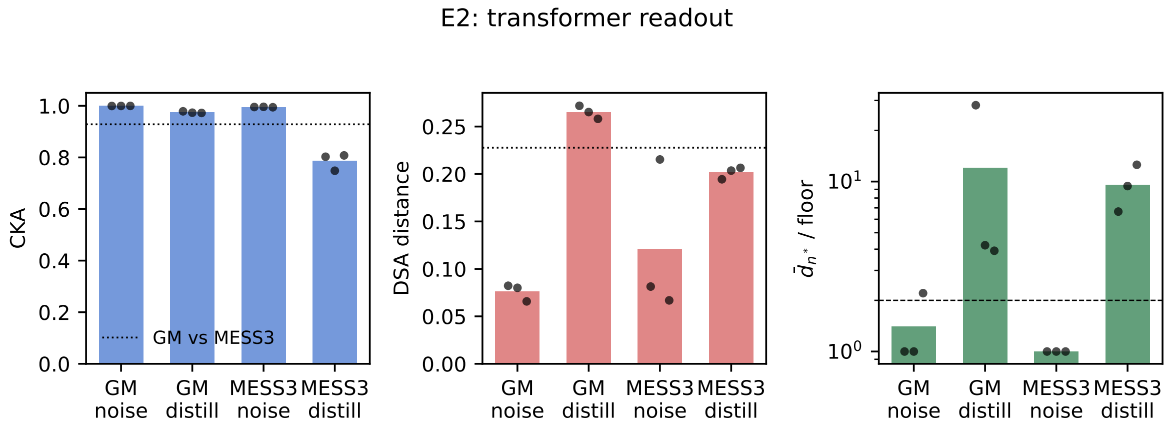

Figure 5: E2 summary: three metrics on transformer pairs (bars = mean over 3 seeds, points = seeds; dotted line = GM-vs-MESS3 cross anchor). $\bar{d}$ shown at the amended plateau as a multiple of the same-process floor.

ordering (noise < distill ≪ cross-task) is consistent on every seed.

Residual noise: the readout hierarchy again. With fresh noise injected into the residual stream at every forward pass, the emitted process is essentially unchanged (separation on 1/3 seeds only; $\bar{d}_1 \leq 0.001$) — and here even DSA stays quiet (0.076 ± 0.007; no recurrent state means the noise does not propagate through fitted dynamics, so the errors-in-variables artifact of §5 largely disappears). The belief readout still separates on every seed ($61 \pm 8\times$ floor): the twin computes visibly noisier conditional probabilities while emitting statistically indistinguishable symbols. No baseline, and no static statistic of the emitted stream, sees this distinction; the process metric at the right readout does.

8 What $\bar{d}$ does not do

No general isomorphism detection (K3). By Ornstein and Weiss [18], nothing finite-data can decide process isomorphism beyond entropy; $\bar{d}$ compares processes in a fixed alphabet. Two models could implement isomorphic dynamics that our readout codes differently — $\bar{d}$ would separate them, correctly, *as readout processes*; and the readout is part of the metric's identity.

Readout dependence is real. The emitted-token readout integrates out internal randomness that the belief readout exposes (Table 1: distillation is a null at one readout and not the other). We regard this as a feature — the two readouts answer behavioral vs. mechanistic versions of "did the process change?" — but it means every $\bar{d}$ claim must name its readout.

The wall is a budget constraint. With 2.1×10^6 symbols we buy $n \lesssim 31.4$ at $h = 2/3$ bits. High-entropy readouts (raw activations, large-vocabulary LLM token streams) are out of reach of consistent $\bar{d}$ estimation at realistic budgets — this is a theorem-shaped limitation [11], and the reason our scope is restricted to low-entropy symbolic readouts. Scaling this method to LLM model-diffing requires entropy-reducing readouts (extracted automata [1, 24], constrained decoding, or task-restricted vocabularies), which we flag as the main open engineering problem.

DSA's defense. DSA answers "same deterministic linearized flow?" — a question $\bar{d}$ cannot answer (it never sees the vector field). On pairs where determinism is the object of interest, DSA is the right tool and $\bar{d}$ is silent. Our results say only: for *process-level* sameness — the version of "same computation" relevant to noise, pruning, distillation, and seed variation at matched outputs — the conjugacy question and the process question come apart, in both directions.

9 Conclusion

We imported a fifty-year-old ergodic-theory metric into model comparison and found that, on the restricted domain where its estimator is honest, it sees what the field's current tools do not: pruned twins (at every fraction), independent seeds, transformer distillates, and noise-injected twins with matched outputs are all different *processes*, invisibly to CKA and DSA; different computations with identical static statistics are separated by two orders of magnitude; recoded models and faithful (GRU) distillates are correctly certified as process-identical. Along the way the baselines' failure modes became data: DSA responds to process noise more strongly than to a change of computation, and rates transformer distillates as more different than cross-task model pairs. The pre-registered gate verdict, its failure, and the two amendments that repair it are all in the repository, with every number machine-generated from committed results.

Reproducibility. Code, pre-registration (`PLAN.md`, committed before results), all results JSONs, and scripts regenerating every number and figure in this paper (`gen_paper_numbers.py`, verified by `verify_regen.py`) are in the project repository. All experiments run on one consumer GPU (RTX 3080, 10 GB).

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